

RIEMANN AND HIS ZETA FUNCTION

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ABSTRACT. In this paper, we will discuss the life of the German mathematician Bernhard Riemann. He was born in 1826 and died in 1866. Although he didn't have many published works, he made numerous contributions to the study of mathematics, especially in geometry, complex analysis, real analysis, and number theory. However, we will mainly discuss his work with the Riemann-Zeta Function and Riemann Hypothesis.

1. BIOGRAPHY

Bernhard Riemann was born in 1826 in Hanover, Germany into a Lutheran family. His father was a Lutheran minister who also taught Riemann until he was ten years old.

In 1842, he attended the Johanneum Gymnasium in Lüneberg, where he began to show a penchant for mathematics. The director of the Gymnasium allowed Riemann to borrow mathematical texts from his personal library, including Legendre's book on number theory, which Riemann read rapidly.

In 1846, he enrolled at the University of Göttingen, where he initially planned to study theology. However, after attending a few mathematics lectures, Riemann decided to study mathematics, where he took courses from Moritz Stern and Carl Friedrich Gauss. Although Gauss didn't notice Riemann's talent, Stern did.

In the spring of 1847, he enrolled at Berlin University, where he studied under famous mathematicians such as Jakob Steiner, Carl Jacobi, Lejeune Dirichlet, and Gotthold Eisenstein. Here he was greatly influenced by Eisenstein, with whom he explored the use of complex variables, and Dirichlet, whose teaching style suited Riemann the best.

In 1849, he returned to University of Göttingen for his Ph.D., where he worked under the guidance of Gauss. During his time here, however, he learned a decent amount of theoretical physics and topology, which helped him write his thesis on the theory of complex variables. Specifically, his thesis, which was submitted in

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1851, discussed what we now call Riemann surfaces, which help others study the global behavior of holomorphic functions ¹.

Afterwards, he was recommended by Gauss for a position as a lecturer in Göttingen, which required that he write a dissertation. His dissertation dealt with representing functions using trigonometric series, which are series of the form

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos(nx) + B_n \sin(nx)).$$

More importantly, however, the dissertation also discussed conditions to determine whether a function could be integrated and the first rigorous definition of a definite integral, the Riemann Integral [2].

In 1859, Riemann was appointed as chair of the mathematics department at Göttingen and elected to the Berlin Academy of Sciences, which required that he submit a paper regarding his most recent research. As a result, he submitted his paper titled "*On the number of primes less than a given magnitude*", which was a landmark paper in analytic number theory. In the span of eight and one-half pages, he discussed the zeta function $\zeta(s)$ and its roots (this is subsequently the Riemann Hypothesis) and contemplated the connection between $\zeta(s)$ and the distribution of prime numbers.

In 1862, he married in Elise Koch, with whom he had one daughter. Unfortunately, a few months thereafter, he contracted tuberculosis, which caused an attack of pleurisy, a condition where a membrane lining the inner side of the chest cavity becomes inflamed. As an attempt to recover over the next few years, he made repeated trips to Italy. Unfortunately, they failed to cure him and he passed away in Italy in June 1866.

After his death, some of his unfinished work was thrown away, but mathematician Richard Dedekind was able to salvage some of his papers and lectures and publish them.

In Section 2 of this paper, we will discuss the Riemann Zeta Function and how his worked diverged from that of Leonhard Euler. In Section 3 of this paper, we will discuss the Riemann Hypothesis.

¹*Holomorphic functions*: A functions is called holomorphic on a region $\mathbf{R}$ if it is complex differentiable at every point in $\mathbf{R}$ [6]

2. INTRODUCTION TO RIEMANN ZETA FUNCTION

Amongst Riemann's many contributions to mathematics, one of his biggest influences was specifically in analytic number theory. Even though he only published one paper in number theory, "*On the number of primes less than a given magnitude*", it was a landmark text in not only the field of number theory, and even had far-reaching impacts in all of mathematics and physics. One of the biggest insights in this seminal paper was his discussion of the Riemann zeta function $\zeta(s)$.

In order to define the Riemann zeta function $\zeta(s)$, we recall the definition of a Dirichlet series. A *Dirichlet series* is defined as a series

$$\sum a(n)e^{-\lambda(n)z},$$

where $a(n), z \in \mathbb{C}$ and $\{\lambda(n)\}$ is a strictly increasing sequence of real numbers [8].

A *zeta function* is defined as a Dirichlet series,

$$F(n) := \sum_{k=1}^{\infty} [f(k)]^n,$$

where $f(k)$ represents the set of zeros of some function [4].

The *Riemann zeta function* $\zeta(s)$, the most common form of the zeta function, is defined as [5]

$$Z(s) := \sum_{k=1}^{\infty} \frac{1}{k^s}. \quad (2.1)$$

Euler looked into $Z(s)$, defined in (2.1), upon discovering the following results:

$$Z(1) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots + \frac{1}{n} + \cdots,$$

which doesn't converge. Another interesting and well-known result is

$$\begin{aligned} Z(2) &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots + \frac{1}{n^2} + \cdots \\ &= \frac{\pi^2}{6}. \end{aligned}$$

A lesser-known, but interesting result is [5]

$$\begin{aligned} Z(4) &= 1 + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \frac{1}{5^4} + \cdots + \frac{1}{n^4} + \cdots \\ &= \frac{\pi^4}{90}. \end{aligned}$$

These discoveries led him to search for patterns in $Z(s)$ for integral value of $s \neq 2, 4$, but it proved difficult to find any sort of meaningful patterns.

Here we must take a quick break to examine another property of the Riemann zeta function. There is a deep connection between prime numbers and the Riemann zeta function.

Theorem 2.1. *Euler's Product Formula connected with Riemann zeta function [7]*

$$Z(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}},$$

where $\mathbb{P}$ represents the set of all prime numbers.

Proof. We can rewrite the expression inside the infinite product to be

$$\begin{aligned} \frac{1}{1 - p^{-s}} &= \frac{1}{1 - \frac{1}{p^s}} \\ &= 1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \frac{1}{p^{4s}} + \dots \end{aligned}$$

As a result, we can rewrite the product as

$$\begin{aligned} \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}} &= \prod_{p \in \mathbb{P}} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \frac{1}{p^{4s}} + \dots \right) \\ &= \left(1 + \frac{1}{2^s} + \frac{1}{2^{2s}} + \dots \right) \left(1 + \frac{1}{3^s} + \frac{1}{3^{2s}} + \dots \right) \left(1 + \frac{1}{5^s} + \frac{1}{5^{2s}} + \dots \right) \dots \\ &= 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{2^{2s}} + \frac{1}{5^s} + \frac{1}{2^{2s}} \cdot \frac{1}{3^s} + \frac{1}{7^s} + \frac{1}{2^{3s}} + \dots \\ &= \sum_{k=1}^{\infty} \frac{1}{k^s} \end{aligned}$$

Since every positive integer can be expressed as a finite product of prime numbers, expanding that infinite product of converging infinite sums will result in the infinite sum defining the Riemann zeta function, as we see above. □

However, there was one flaw with $Z(s)$, namely that it was undefined for any negative value of s . Riemann found a better version for the zeta function, which we will look at in the next section.

3. RIEMANN HYPOTHESIS

Riemann also picked up work on the zeta function, but rewrote it so that it could handle negative values of s . His work resulted in the Riemann zeta function $\zeta(s)$ defined as follows:

Theorem 3.1. *Riemann zeta function [5]*

For $s \in \mathbb{C}_{\neq 1}$,

$$\zeta(s) = \frac{1}{1 - 2^{1-s}} \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \sum_{k=0}^n (-1)^k \binom{n}{k} (k+1)^{-s},$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

Interestingly enough, for all $s \geq 1$, $\zeta(s) = Z(s)$. However, $\zeta(s)$ had an expanded domain to include all $s \in \mathbb{C}_{\neq 1}$.

Riemann looked for interesting properties of this new $\zeta(s)$, most notably the roots of this function. During this exploration, it became clear that there were infinitely many "trivial" solutions, namely $s = -2k, k \in \mathbb{N}[3]$. However, he noticed that non-trivial solutions followed a interesting pattern, which has come to be known as the Riemann hypothesis.

Conjecture 3.2. *Riemann's Hypothesis [7]*

If $\zeta(s) = 0$, then $s = \frac{1}{2} + iy, y \in \mathbb{R}_{\neq 0}$ or $s = -2k, k \in \mathbb{Z}$.

Unfortunately, this problem has remained a conjecture without any known counterexample, but it has been yet to be proven true for all values s of the given form. ²

This hypothesis is very important for several reasons, chief amongst which is Gauss's Prime Number Theorem.

Theorem 3.3. *Prime Number Theorem [4]*

For $x > 1$, the number of primes less than x , $P(x)$, is approximately equal to

$$Li(x) = \int_2^x \frac{dt}{\ln t} \approx \frac{x}{\ln x}.$$

The natural question to explore becomes finding the error term of $Li(x)$. If the Riemann hypothesis is assumed to be true, then $|Li(x) - P(x)| \leq \sqrt{x} \ln x$ [1]. Interestingly enough, the converse is true as well: if $|Li(x) - P(x)| \leq \sqrt{x} \ln x$, then the Riemann hypothesis is correct, presenting perhaps another way to find this elusive proof.



²Proving this conjecture, a Millenium Prize Problem, can fetch a person US \$ 1 Million [9].

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